Cross-Model Embedding Adapters: Forward and Backward Compatibility Without Re-Embedding

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Abstract

We evaluate five alignment methods for cross-model embedding spaces on 12 directed model pairs (both forward and backward compatibility) across 13 NanoBEIR retrieval tasks. With 5000 calibration samples, orthogonal Procrustes (a single SVD) wins 9 of 12 pairs. A 3-layer MLP trained with InfoNCE is consistently the worst method, confirming that the alignment problem is fundamentally linear. KRR-RBF collapses on high-dimensional targets but remains competitive on low-dimensional ones. The key predictor of alignment success is CKA similarity between source and target spaces (r=0.89): pairs above CKA $>$ 0.9 align well via rotation; pairs below 0.8 resist alignment regardless of adapter complexity. Cross-model compatibility can be decided by a single CKA measurement, and when alignment is feasible, a single SVD suffices.

1 Introduction

Switching embedding models in a production retrieval system breaks cross-model compatibility: queries from one model cannot retrieve documents encoded by another. A lightweight adapter that maps between embedding spaces restores forward and backward compatibility without re-embedding the corpus. We evaluate five such adapters, from a closed-form SVD rotation to a 3-layer MLP, on 12 directed model pairs across 13 NanoBEIR retrieval tasks.

No single adapter dominates all pairs, but the simplest one comes closest. Across four Jina embedding models (v3, v4, v5-nano, v5-small) spanning 768 to 2048 dimensions, orthogonal Procrustes (a closed-form rotation requiring one SVD) wins 9 of 12 directed model pairs on downstream retrieval with 5000 calibration samples. A nonlinear MLP adapter is consistently the worst method, while RidgeCCA or SGD-Linear win the remaining three pairs, two of which involve dimensionality reduction from a 2048d source. The deciding factor is geometric similarity: CKA between embedding spaces predicts alignment quality (r=0.89, p=0.019), independent of model lineage or parameter count.

These findings complement UniCon [Sui et al., 2026], which showed that contrastive alignment objectives admit closed-form kernel solutions. We extend this perspective empirically: with 5000 calibration samples, the closed-form Procrustes solution is the most robust choice, but CCA-based subspace projection outperforms it when the source dimensionality exceeds the target. Nonlinear extensions (KRR-RBF) fail on high-dimensional targets but remain viable for low-dimensional ones. Prior work on cross-lingual word embedding alignment via Procrustes [Conneau et al., 2018] showed similar patterns in lower-dimensional spaces; our results confirm this holds for modern high-dimensional sentence encoders.

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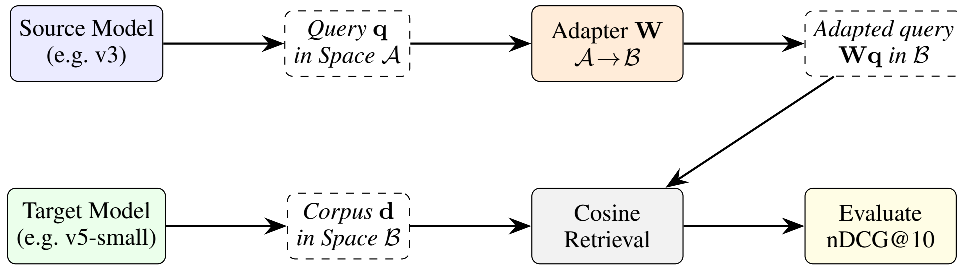

Figure 1: Adapter pipeline: $\mathbf{q} \in \mathcal{A} \xrightarrow{f} \mathcal{B}$, then cosine retrieval against $\mathbf{d} \in \mathcal{B}$.

Notation. We denote embedding matrices as $\mathbf{X} \in \mathbb{R}^{n \times d_s}$, $\mathbf{Y} \in \mathbb{R}^{n \times d_t}$, query and document vectors as $\mathbf{q} \in \mathbb{R}^{d_s}$, $\mathbf{d} \in \mathbb{R}^{d_t}$, adapter matrices as $\mathbf{R}$ (orthogonal) or $\mathbf{W}$ (learned), and embedding spaces as $\mathcal{A}$, $\mathcal{B}$.

Goal. Given a *source* model producing queries $\mathbf{q} \in \mathcal{A} \subseteq \mathbb{R}^{d_s}$ and a *target* model producing corpus embeddings $\mathbf{d} \in \mathcal{B} \subseteq \mathbb{R}^{d_t}$, learn an adapter $f : \mathcal{A} \to \mathcal{B}$ such that cross-model retrieval $\text{sim}(f(\mathbf{q}), \mathbf{d})$ approaches the quality of *target*-model-only retrieval (Figure 1).

No Alignment. $f(\mathbf{q}) = \text{pad}(\mathbf{q}) \in \mathbb{R}^{d_t}$, zero-padding when $d_s < d_t$ or truncating when $d_s > d_t$.

Procrustes. $f(\mathbf{q}) = \mathbf{R}^*\mathbf{q}$, where $\mathbf{R}^*$ is a closed-form orthogonal rotation:

$$\mathbf{R}^* = \arg\min_{\mathbf{R}} \|\mathbf{X}\mathbf{R} - \mathbf{Y}\|_F \quad \text{s.t.} \quad \mathbf{R}^\top \mathbf{R} = \mathbf{I}$$

Solution: let $\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top = \text{SVD}(\mathbf{X}^\top \mathbf{Y})$, then $\mathbf{R}^* = \mathbf{U}\mathbf{V}^\top$.

RidgeCCA. $f(\mathbf{q}) = \mathbf{A}^*\mathbf{q}$, $g(\mathbf{d}) = \mathbf{B}^*\mathbf{d}$, bilateral projection into a shared k-dimensional subspace (k=128):

$$\mathbf{A}^*, \mathbf{B}^* = \arg\max_{\mathbf{A},\mathbf{B}} \text{corr}(\mathbf{X}\mathbf{A},\ \mathbf{Y}\mathbf{B}) \quad \text{with ridge regularization } \lambda{=}10^{-4}$$

Retrieval uses $\text{sim}(f(\mathbf{q}),\ g(\mathbf{d}))$ in the shared subspace.

KRR-RBF. Kernel ridge regression with RBF kernel $k(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma^2)$, bandwidth σ set by the median heuristic:

$$f(\mathbf{q}) = \mathbf{K}_{\mathbf{q},\mathbf{X}}(\mathbf{K}_{\mathbf{X},\mathbf{X}} + \lambda \mathbf{I})^{-1}\mathbf{Y}$$

SGD-Linear-InfoNCE. $f(\mathbf{q}) = \mathbf{W}\mathbf{q}$, where $\mathbf{W} \in \mathbb{R}^{d_t \times d_s}$ is trained with InfoNCE loss:

$$\mathcal{L} = -\frac{1}{n}\sum_{i=1}^{n} \log \frac{\exp(\cos(\mathbf{W}\mathbf{x}_i,\ \mathbf{y}_i)/\tau)}{\sum_{j=1}^{n} \exp(\cos(\mathbf{W}\mathbf{x}_i,\ \mathbf{y}_j)/\tau)}$$

with τ=0.05, batch size 256, 500 epochs, initialized from the least-squares solution.

SGD-MLP-InfoNCE. Same loss as above, but $f(\mathbf{q}) = \mathbf{W}_3\, \sigma(\mathbf{W}_2\, \sigma(\mathbf{W}_1 \mathbf{q}))$, where σ = ReLU, $\mathbf{W}_1 \in \mathbb{R}^{512 \times d_s}$, $\mathbf{W}_2 \in \mathbb{R}^{512 \times 512}$, $\mathbf{W}_3 \in \mathbb{R}^{d_t \times 512}$. Trained with cosine learning rate schedule, 500 epochs.

We design our experiments to answer three questions:

RQ1 (Closed-form vs. iterative). Does the UniCon theoretical equivalence hold on real retrieval benchmarks with 5000 calibration samples?

RQ2 (Nonlinear extensions). Do nonlinear kernel extensions (KRR-RBF) or subspace methods (CCA) improve over Procrustes, or does additional flexibility hurt under data-scarce alignment?

2 Experimental Setup

We evaluate four text embedding models with different architectures and dimensionalities (Table 1), yielding 12 directed model pairs (6 forward + 6 backward). All models are instruction-tuned; the 10 asymmetric retrieval tasks use `retrieval.query`/`retrieval.passage` prefixes, while the 3 symmetric tasks (QuoraRetrieval, SCIDOCS, ArguAna) use `text-matching` for both queries and documents. We report nDCG@10 averaged across all 13 NanoBEIR tasks. All adapters are trained on 5000 calibration texts (2500 query + 2500 passage embeddings from each model), sampled from AG News with zero overlap with NanoBEIR.

Table 1: Embedding models used in our experiments, sorted by native retrieval performance.

Model	Dim	Native nDCG@10	Notes
jina-embeddings-v5-text-small	1024	0.671	mid-size
jina-embeddings-v5-text-nano	768	0.667	lightweight
jina-embeddings-v4	2048	0.647	multimodal
jina-embeddings-v3	1024	0.634	previous generation

We measure inter-model representational similarity using linear Centered Kernel Alignment (CKA) [Kornblith et al., 2019], computed on the 5000 calibration embeddings. Given two embedding matrices $\mathbf{X} \in \mathbb{R}^{n \times d_1}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d_2}$ from the same n inputs, linear CKA is:

$$\mathrm{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{Y}^\top \mathbf{X}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \cdot \|\mathbf{Y}^\top \mathbf{Y}\|_F}$$

CKA is invariant to dimensionality and rotation, yielding a score in $[0, 1]$ where 1 indicates identical representational geometry.

3 Results

3.1 Main Results

Table 2 presents all 12 directed model pairs, split into forward compatibility (source nDCG@10 $<$ target) and backward compatibility (source nDCG@10 $>$ target). Figure 2 visualizes the full result matrix.

Table 2: Average nDCG@10 across 13 NanoBEIR tasks for all 12 directed model pairs. Best adapter per pair in **bold**. No-alignment baseline ≈0.003 for all pairs. Calibration: 5000 samples.

Pair	CKA	Native Src	Procrustes	SGD-Linear	KRR-RBF	RidgeCCA	SGD-MLP
Forward compatibility (source nDCG@10 $<$ target)							
nano→small	0.960	0.667	**0.666**	0.643	0.661	0.647	0.580
v3→small	0.776	0.634	**0.548**	0.536	0.527	0.546	0.459
v3→nano	0.770	0.634	**0.549**	0.523	0.536	0.539	0.466
v4→small	0.770	0.647	0.586	**0.594**	0.574	0.593	0.487
v4→nano	0.759	0.647	**0.587**	0.582	0.572	0.585	0.488
v3→v4	0.730	0.634	**0.540**	0.501	0.288	0.535	0.422
Backward compatibility (source nDCG@10 $>$ target)							
small→nano	0.960	0.671	**0.660**	0.636	0.658	0.646	0.576
small→v4	0.770	0.671	**0.612**	0.594	0.420	0.606	0.465
small→v3	0.776	0.671	0.558	0.555	0.564	**0.568**	0.469
nano→v3	0.770	0.667	**0.559**	0.558	0.553	0.565	0.459
nano→v4	0.759	0.667	**0.613**	0.584	0.412	0.608	0.456
v4→v3	0.730	0.647	0.434	**0.535**	0.504	0.521	0.430

Figure 2: Adapter performance (% of native source nDCG@10) across 12 directed model pairs and 5 adapter methods. Procrustes dominates when source dimensionality $\leq$ target, but CCA and SGD win when mapping from higher to lower dimensions.

3.2 Key Findings

Finding 1 (RQ1): Procrustes dominates. With 5000 calibration samples, Procrustes wins 9 of 12 directed pairs. The nano→small pair (CKA=0.96) reaches 0.666 nDCG@10, within 0.001 of the native v5-nano baseline (0.667).

Two of the three exceptions involve v4 (2048d) as source: v4→v3 is won by SGD-Linear (0.535 vs. Procrustes 0.434) and v4→small by SGD-Linear (0.594 vs. 0.586), since orthogonal rotation cannot compress 2048 dimensions into a lower-dimensional space. The third, small→v3 (both 1024d), is won narrowly by RidgeCCA (0.568 vs. 0.558).

Finding 2 (RQ2): Nonlinear adapters underperform linear ones. SGD-MLP-InfoNCE is the worst method across all 12 pairs, consistently 10 to 20% below Procrustes. With 5000 calibration samples, the MLP overfits: its additional parameters are a liability, not an asset. This confirms that cross-model alignment is fundamentally linear when embedding spaces share similar geometry.

KRR-RBF shows dimensional sensitivity: competitive on low-dimensional targets (nano→v3: 0.553, v4→v3: 0.504) but collapses when mapping to 2048d targets (v3→v4: 0.288, nano→v4: 0.412, small→v4: 0.420). With only 5000 training points in 2048 dimensions, the RBF kernel memorizes rather than generalizes.

4 Discussion

Practical implications. For production systems requiring cross-model compatibility, our results suggest a decision process: (1) Compute CKA between old and new model embeddings on a small sample. (2) If CKA > 0.9, a Procrustes rotation trained on $\sim$5000 paired embeddings bridges the models with minimal quality loss. If CKA < 0.8, no lightweight adapter recovers acceptable retrieval quality; re-embedding is necessary. (3) When the source model has higher dimensionality than the target, consider RidgeCCA: its shared-subspace projection handles dimensionality reduction more gracefully than orthogonal rotation.

Connection to UniCon. Our findings complement the theoretical analysis of Sui et al. [2026], who showed that contrastive alignment objectives admit closed-form solutions via spectral decomposition. Procrustes outperforming SGD-InfoNCE validates their insight: when alignment is an orthogonal mapping, the closed-form solution is both faster and more accurate than iterative optimization on small datasets. RidgeCCA's advantage on high-to-low dimensional pairs further supports this: bilateral spectral projections outperform unilateral rotation when dimensionality reduction is inherent to the task. KRR-RBF's collapse on high-dimensional targets confirms that kernelized solutions require sufficient training data relative to target dimensionality.

Limitations. Our study uses NanoBEIR (50 queries per task), which introduces variance in per-task nDCG@10 estimates. We evaluate only Jina embedding models; other model families (e.g., OpenAI, Cohere) may exhibit different CKA patterns.

5 Conclusion

Across 12 directed model pairs, 5 adapter methods, and 13 retrieval tasks, Procrustes wins 9 of 12 pairs with 5000 calibration samples. The nonlinear MLP adapter is consistently the worst, confirming that cross-model alignment is a linear problem when embedding spaces share similar geometry. CKA is the primary predictor of alignment success (r=0.89), and the only systematic exception to Procrustes dominance occurs when the source has higher dimensionality than the target.

References

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